

EXERCISE-1

SCENARIO-1

```
/*SCENARIO1*/
BEGIN
  FOR cust IN (
    SELECT CustomerID
    FROM Customers
    WHERE Age > 60
  )
  LOOP
    UPDATE Loans
    SET InterestRate = GREATEST(InterestRate - 1, 0)
    WHERE CustomerID = cust.CustomerID;
  END LOOP;

  COMMIT;
  DBMS_OUTPUT.PUT_LINE('Interest rate discount applied successfully.');
```

OUTPUT SCENARIO-1

```
SQL> BEGIN
      FOR cust IN (
        SELECT CustomerID
        FROM Customers...
Show more...

Interest rate discount applied successfully.

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.012
```

SCENARIO-2

```
35  /*SCEANRIO 2:Promote Customers to VIP Status Based on Balance*/
36  BEGIN
37      FOR cust IN (
38          SELECT CustomerID
39          FROM Customers
40          WHERE Balance > 10000
41      )
42      LOOP
43          UPDATE Customers
44          SET IsVIP = 'TRUE'
45          WHERE CustomerID = cust.CustomerID;
46      END LOOP;
47
48      COMMIT;
49      DBMS_OUTPUT.PUT_LINE('VIP status updated successfully.');
```

OUTPUT OF SCENARIO-2

Query result Script output DBMS output Explain Plan SQL history					
Download Execution time: 0.004 seconds					
	CUSTOMERID	NAME	AGE	BALANCE	ISVIP
1	1	John	65	15000	TRUE
2	2	Smith	45	8000	FALSE
3	3	David	70	12000	TRUE
4	4	Alice	30	9000	FALSE

```
SQL> SELECT LoanID, CustomerID, DueDate
      FROM Loans

LOANID CUSTOMERID DUEDATE
-----
101      1          07/14/2026, 11:04:21 AM
102      2          08/03/2026, 11:04:21 AM
103      3          07/04/2026, 11:04:21 AM

Elapsed: 00:00:00.006
3 rows selected.
```

SCENARIO -3

```
/*Scenario 3: Send Reminders to Customers Whose Loans Are Due Within the Next 30 Days*/
BEGIN
  FOR loan_rec IN (
    SELECT c.Name,
           l.LoanID,
           l.DueDate
    FROM Customers c
    JOIN Loans l
    ON c.CustomerID = l.CustomerID
    WHERE l.DueDate BETWEEN SYSDATE AND SYSDATE + 30
  )
  LOOP
    DBMS_OUTPUT.PUT_LINE(
      'Reminder: Dear ' || loan_rec.Name ||
      ', your Loan ID ' || loan_rec.LoanID ||
      ' is due on ' || TO_CHAR(loan_rec.DueDate, 'DD-MON-YYYY')
    );
  END LOOP;
END;
/

SELECT LoanID, CustomerID, DueDate
FROM Loans;
```

OUTPUT-3

Query result Script output DBMS output Explain Plan SQL history				
Download Execution time: 0.002 seconds				
	LOANID	CUSTOMERID	DUEDATE	
1	101	1	7/14/2026, 5:34:21	
2	102	2	8/3/2026, 5:34:21 A	
3	103	3	7/4/2026, 5:34:21 A	

Query result Script output DBMS output Explain Plan SQL history				
SQL> BEGIN FOR loan_rec IN (SELECT c.Name, l.LoanID,... Show more... Reminder: Dear John, your Loan ID 101 is due on 14-JUL-2026 Reminder: Dear David, your Loan ID 103 is due on 04-JUL-2026 PL/SQL procedure successfully completed. Elapsed: 00:00:00.006				

EXERCISE-2

EXERCISE_2.sql* ▶ ⌕ ↺ ⌕ Aa ⌕				
1	/*Scenario 1: SafeTransferFunds*/			
2	CREATE OR REPLACE PROCEDURE SafeTransferFunds(			
3	p_fromAccount IN NUMBER,			
4	p_toAccount IN NUMBER,			
5	p_amount IN NUMBER			
6	)			
7	IS			
8	v_balance NUMBER;			
9	BEGIN			
10	SELECT Balance			
11	INTO v_balance			
12	FROM Accounts			
13	WHERE AccountID = p_fromAccount;			
14				
15	IF v_balance < p_amount THEN			
16	RAISE_APPLICATION_ERROR(-20001, 'Insufficient Funds');			
17	END IF;			
18				
19	UPDATE Accounts			
20	SET Balance = Balance - p_amount			
21	WHERE AccountID = p_fromAccount;			
22				
23	UPDATE Accounts			
24	SET Balance = Balance + p_amount			
25	WHERE AccountID = p_toAccount;			
26				

```

9      BEGIN
18
19      UPDATE Accounts
20      SET Balance = Balance - p_amount
21      WHERE AccountID = p_fromAccount;
22
23      UPDATE Accounts
24      SET Balance = Balance + p_amount
25      WHERE AccountID = p_toAccount;
26
27      COMMIT;
28
29      DBMS_OUTPUT.PUT_LINE('Funds transferred successfully.');
```

```

31 EXCEPTION
32     WHEN OTHERS THEN
33         ROLLBACK;
34         DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);
35 END;
36 /
37 BEGIN
38     SafeTransferFunds(101,102,1000);
39 END;
40 /
```

OUTPUT

```

SQL> BEGIN
      SafeTransferFunds(101,102,1000);
      END;

Funds transferred successfully.

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.021
```

SCENARIO-2

```

EXERCISE_2.sql*
42  /*Scenario 2: UpdateSalary*/
43  > CREATE OR REPLACE PROCEDURE UpdateSalary(...
46  )
47  IS
48  BEGIN
49      UPDATE Employees
50      SET Salary = Salary + (Salary * p_percent/100)
51      WHERE EmployeeID = p_empid;
52
53      IF SQL%ROWCOUNT = 0 THEN
54          RAISE NO_DATA_FOUND;
55      END IF;
56
57      COMMIT;
58
59      DBMS_OUTPUT.PUT_LINE('Salary updated successfully.');
```

```

61 EXCEPTION
62     WHEN NO_DATA_FOUND THEN
63         DBMS_OUTPUT.PUT_LINE('Error: Employee ID does not exist.');
```

```

64 END;
65 /
66 BEGIN
67     UpdateSalary(1,10);
68 END;
69 /
```

OUTPUT

```
SQL> BEGIN
      UpdateSalary(1,10);
      END;

Salary updated successfully.

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.013
```

SCENARIO-3

```
70
71 /*Scenario 3: AddNewCustomer*/
72 CREATE OR REPLACE PROCEDURE AddNewCustomer(
73     p_id IN NUMBER,
74     p_name IN VARCHAR2,
75     p_age IN NUMBER,
76     p_balance IN NUMBER,
77     p_vip IN VARCHAR2
78 )
79 IS
80 BEGIN
81     INSERT INTO Customers
82     VALUES (p_id,p_name,p_age,p_balance,p_vip);
83
84     COMMIT;
85
86     DBMS_OUTPUT.PUT_LINE('Customer added successfully.');
```

```
SQL> BEGIN
      AddNewCustomer(4, 'Alice', 30, 9000, 'FALSE');
      END;

Error: Customer ID already exists.

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.012
```

EXERCISE-3

```
Exercise_3.sql*  ▶ ⏏ ⏏ ⏏ ⏏ ⏏ Aa ⏏  
1 CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest  
2 IS  
3 BEGIN  
4     UPDATE Accounts  
5     SET Balance = Balance + (Balance * 0.01);  
6  
7     COMMIT;  
8  
9     DBMS_OUTPUT.PUT_LINE('Monthly interest processed successfully.');
```

```
SQL> CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest  
      IS  
      BEGIN  
          UPDATE Accounts...  
Show more...
```

Procedure PROCESSMONTHLYINTEREST compiled

Elapsed: 00:00:00.007

```
SQL> BEGIN  
      ProcessMonthlyInterest;  
      END;
```

Monthly interest processed successfully.

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.009

SCENARIO-2

```
Exercise_3.sql*
26 CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(
27     p_department IN VARCHAR2,
28     p_bonus_percent IN NUMBER
29 )
30 IS
31 BEGIN
32     UPDATE Employees
33     SET Salary = Salary + (Salary * p_bonus_percent / 100)
34     WHERE Department = p_department;
35
36     COMMIT;
37
38     DBMS_OUTPUT.PUT_LINE('Employee bonus updated successfully.');
```

```
SQL> CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(
      p_department IN VARCHAR2,
      p_bonus_percent IN NUMBER
    )...
Show more...
```

Procedure UPDATEEMPLOYEEBONUS compiled

Elapsed: 00:00:00.005

```
SQL> BEGIN
      UpdateEmployeeBonus('IT',10);
```

```
SQL> BEGIN
      UpdateEmployeeBonus('IT',10);
END;
```

Employee bonus updated successfully.

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.005

SCENARIO-3

```
SQL> CREATE OR REPLACE PROCEDURE TransferFunds(
      p_fromAccount IN NUMBER,
      p_toAccount IN NUMBER,
      p_amount IN NUMBER...
Show more...
```

Procedure TRANSFERFUNDS compiled

Elapsed: 00:00:00.006

```
SQL> BEGIN
      TransferFunds(101,102,1000);
```

Elapsed: 00:00:00.006

```
SQL> BEGIN
      TransferFunds(101,102,1000);
      END;
```

Funds transferred successfully.

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.018

Exercise_3.sql*

```
45
46 CREATE OR REPLACE PROCEDURE TransferFunds(
47     p_fromAccount IN NUMBER,
48     p_toAccount IN NUMBER,
49     p_amount IN NUMBER
50 )
51 IS
52     v_balance NUMBER;
53 BEGIN
54     SELECT Balance
55     INTO v_balance
56     FROM Accounts
57     WHERE AccountID = p_fromAccount;
58
59     IF v_balance < p_amount THEN
60         DBMS_OUTPUT.PUT_LINE('Insufficient balance.');
```

```
61         RETURN;
62     END IF;
63
64     UPDATE Accounts
65     SET Balance = Balance - p_amount
66     WHERE AccountID = p_fromAccount;
67
68     UPDATE Accounts
69     SET Balance = Balance + p_amount
70     WHERE AccountID = p_toAccount;
71
72     COMMIT;
73
74     DBMS_OUTPUT.PUT_LINE('Funds transferred successfully.');
```